

Leticia Mattos Da Silva

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Education

Massachusetts Institute of Technology	Sept. 2021 - <i>present</i>
Ph.D. in Computer Science, advised by Justin Solomon	
Massachusetts Institute of Technology	June 2023
S.M. in Computer Science, advised by Justin Solomon	
University of California, Los Angeles	March 2021
B.S. in Mathematics <i>cum laude</i>	

Research Positions

Guest Researcher	Aug. 2024 - March 2025
Flatiron Institute, Simons Foundation, New York, NY	
Pre-Doctoral Researcher	May - Aug. 2024
Flatiron Institute, Simons Foundation, New York, NY	
Research Scientist Intern	May - Aug. 2022
Adobe Inc., San Jose, CA (Remote)	
Research Intern	July - Aug. 2021
Fields Institute for Research in Mathematical Sciences, Toronto, ON (Remote)	
Research Intern	July - Aug. 2020
Fields Institute for Research in Mathematical Sciences, Toronto, ON (Remote)	
Student Researcher	June - Aug. 2019
SMALL Research Experience for Undergraduates, Williamstown, MA	

Honors and Awards

WiGRAPH Rising Stars in Computer Graphics	2025
Oberwolfach Leibniz Graduate Student (OWLG)	2025
MathWorks Engineering Fellowship	2024
EECS MathWorks Fellowship	2023
Schwarzman College of Computing Fellowship funded by Google	2022
Google Lime Scholarship <i>semi-finalist</i>	2022
Distinguished Graduate Fellowship in Electrical Engineering and Computer Science	2021

Will Rogers Memorial Scholarship	2021
De Anza Physical Sciences, Mathematics, and Engineering Scholarship	2017

Publications

4. Variational Elastodynamic Simulation Leticia Mattos Da Silva , Silvia Sellán, Natalia Pacheco-Tallaj, Justin Solomon ACM SIGGRAPH 2025	2025
3. Mirror Bridges Between Probability Measures Leticia Mattos Da Silva , Silvia Sellán, Francisco Vargas, Justin Solomon Under Peer Review	2025
2. A Framework for Solving Parabolic Partial Differential Equations on Discrete Domains Leticia Mattos Da Silva , Oded Stein, Justin Solomon ACM Transactions on Graphics	2024
1. Breaking Good: Fracture Modes for Realtime Destruction Silvia Sellán, Jack Luong, Leticia Mattos Da Silva , Aravind Ramakrishnan, Yuchuan Yang, Alec Jacobson ACM Transactions on Graphics	2022

Invited Talks

Title TBA Visual Computing Seminar, MIT	TBA
A Framework for Solving Parabolic PDE on Discrete Domains Visgraf Seminar, IMPA — Instituto de Matemática Pura e Aplicada, Recording	March 2025
Tackling Nonlinear and Nonconvex Challenges with Geometric Algorithms Computer Graphics Seminar, ISTA, <i>hosted by Samara Ren</i>	Feb. 2025
Geometric Algorithms, Stochasticity and the Fokker-Planck Equation Oberwolfach Workshop <i>Mathematical Imaging and Surface Processing</i>	Feb. 2025
The Differential Equations Multiverse: From PDE to SDE in Computer Graphics Computational Science and Engineering Community Seminar, MIT	Nov. 2024
Towards Broader Parabolic PDE Solvers in Computer Graphics PAB Seminar, The University of Edinburgh, <i>hosted by Amir Vaxman</i>	Sept. 2024
An Iterative High-Order Method for PDE on Moving Surfaces Geometric Computing Lab, New York University, <i>hosted by Daniele Panozzo</i>	Aug. 2024
Variational Characterizations of Nonlinear PDE on Geometric Domains The Scientific AI Group, University of Texas at Austin, <i>hosted by Richard Tsai</i>	May 2024

Contributed Talks

Variational Elastodynamic Simulation SIGGRAPH, Vancouver, BC, Canada	Aug. 2025
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A Framework for Solving Parabolic PDE on Discrete Domains 31st Annual Fall Workshop on Computational Geometry, Tufts University	Nov. 2024
A Framework for Solving Parabolic PDE on Discrete Domains SIGGRAPH, Denver, CO	July 2024
Log of Heat Equation Graduate Women in Course 6 Research Summit, MIT	Nov. 2021
Fracture Harmonics Women in Math Research Night, University of California, Los Angeles	May 2021
When Bands Play in Random Matrix Theory Young Mathematicians Conference, Columbus, OH, with Charles Devlin VI	Aug. 2019

Other Presentations

Convex Relaxation Strategies for Graphics and Geometry Processing <i>Tutorial Course</i> , Eurographics Symposium on Geometry Processing (SGP), Bilbao, Spain	July 2025
Perron's Method for the Landau Equation <i>Poster</i> , ICMS Workshop, Edinburgh, UK	Sept. 2024
An Iterative High-Order Method for PDE on Moving Surfaces <i>Poster</i> , Flatiron Institute, Simons Foundation, New York, NY	Aug. 2024
Solving Parabolic PDE with MATLAB <i>Poster</i> , MathWorks Day, MIT	April 2024
Solving Parabolic PDE on Discrete Domains <i>Poster</i> , ICERM Workshop <i>PDEs and Geometry: Numerical Aspects</i> , Brown University	March 2024

Academic Service and Organizing

Member of the Graduate Student Advisory Group School of Engineering, MIT	Sept. 2024 - <i>present</i>
Co-Organizer of the New England Symposium on Graphics MIT	April 2025
<i>Flipping Failure</i> Advisory Board Teaching and Learning Lab, MIT	Sept. 2023 - May 2024
Organizer of <i>EECS Culture: Today and Tomorrow</i> Event Visit Days, EECS Department, MIT	March 2024
Graduate Applications Reviewer EECS Department, MIT	Jan. 2024
Executive Member of Diversity, Equity, and Inclusion EECS Graduate Student Association, MIT	Jan. 2023 - Dec. 2023
Disability Justice Council Representative Graduate Student Council, MIT	Sept. 2021 - June 2023

Executive Member of New Student Orientation EECS Graduate Student Association, MIT	Jan. 2022 - Dec. 2022
Student Volunteer Eurographics/ACM SIGGRAPH Symposium on Geometry Processing (SGP)	July 2021
Volunteer for Transfer Student Welcome Undergraduate Admission, University of California, Los Angeles	May 2021
Co-organizer of Tutor Training Workshops Math Performance Success Program, De Anza College	Sept. 2017 - June 2018

Outreach

Organizer of Session on Graduate School Applications Summer Geometry Initiative, MIT	Summer 2021, 2023, 2024, 2025
Mentor for the Graduate Assistance Application Program (GAAP) EECS Department, MIT	Fall 2021, 2022, 2024
Volunteer for Graduate Recruitment Effort The Society of Hispanic Engineers (SHPE) National Convention, Salt Lake City, UT	Nov. 2023
Volunteer for the Student Transfer Outreach and Mentor Program (STOMP) Undergraduate Admission, University of California, Los Angeles	April 2021
Co-organizer of Outreach Program <i>Noches de Ciencias</i> Society of Latino Engineers and Scientists, University of California, Los Angeles	Sept. 2019 - March 2020

Mentoring

Ryan Dong, Undergraduate Researcher, UROP, MIT	Spring 2025
Ayyoob Berhane, Undergraduate Researcher, SuperUROP Scholar, MIT	AY 2024 - 2025
Akshata Tiwari, Undergraduate Researcher, UROP, MIT	Indep. Activities Period 2024
Francisco Unai Caja López, Summer Geometry Institute Fellow	Summer 2023
Anna Cole Percy, Summer Geometry Institute Fellow	Summer 2023
Matheus da Silva Araujo, Summer Geometry Institute Fellow	Summer 2023
Hossam Mohamed Saeed Gize, Summer Geometry Institute Fellow	Summer 2023
Dora Zhao, Graduate Application Mentee, <i>now at Stanford</i>	Fall 2022
Olga Guţan, Graduate Application Mentee, <i>now at CMU</i>	Fall 2021

Teaching Positions

Applied Numerical Algorithms Teaching Assistant, MIT	Fall 2025
Progressive Mesh Compression Teaching Assistant, Summer Geometry Institute, MIT, <i>supervised by Paul Kry</i>	Summer 2023

First-Year Engineering Transition Bridge: Calculus
Instructor, University of California, Los Angeles

Summer 2019, 2020

Introduction to Engineering Disciplines
Teaching Assistant, University of California, Los Angeles

Fall 2019

Intermediate Algebra, Precalculus, Integrated Statistics, ...
Teaching Assistant, De Anza College

Fall 2016 - Spring 2018